

Anion-Dependent Molecular Doping and Charge Transport in Ferric Salt-Doped P3HT for Thermoelectric Application

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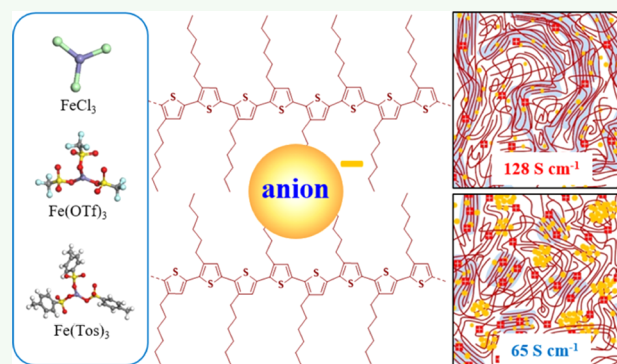
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Supporting Information

ABSTRACT: The effect of different iron (III) dopants on the doping process and charge transport properties based on a poly(3-hexylthiophene) (P3HT) film was investigated. It is found that the doping level is dependent on not only the driving force for charge transfer but also the miscibility between a polymer and a dopant, while the mobile carrier transport is significantly controlled by the microstructure upon doping. A high electrical conductivity (128 S cm^{-1}) is obtained for a FeCl_3 -doped P3HT film among three different doped P3HT combinations, although a low doping level is observed in this film. In contrast, a highest doping level but a low electrical conductivity (65 S cm^{-1}) is achieved for $\text{Fe}(\text{OTf})_3$ -doped P3HT. Another ferric salt with a larger size anion and strong oxidation ability, $\text{Fe}(\text{Tos})_3$, endows both much low doping level and low electrical conductivity (9 S cm^{-1}). Grazing-incidence wide-angle X-ray scattering (GIWAXS) shows that a much stronger π – π stacking of P3HT and larger crystalline domains may exist in $\text{Fe}(\text{OTf})_3$ -doped P3HT compared with those of FeCl_3 -doped P3HT. However, Hall-effect measurements show that the high electrical conductivity of FeCl_3 -doped P3HT is mainly attributed to higher carrier mobility. Temperature-dependent conductivity experiments demonstrate that smaller activation energy for carrier transport is needed for a FeCl_3 -doped P3HT film. These results indicate that smooth and continuous transport paths are formed in a FeCl_3 -doped film, contributing to high carrier mobility while discrete domains in $\text{Fe}(\text{OTf})_3$ -doped film hamper the carrier transport. A prototype device with a five-leg FeCl_3 -doped P3HT film connected with a silver paste was fabricated. The measured maximum output power is about 4.64 nW at the temperature difference of 23.3 K. Our results suggest that the interaction between dopant anions and polymer chains is crucial for high electrical conductivity by improving morphologies to achieve ionized carriers' transfer into much mobile carriers.

KEYWORDS: chemical doping, polymer semiconductor, charge transport, ferric salts, thermoelectric materials



INTRODUCTION

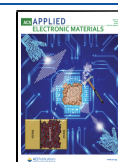
Semiconducting polymers can be solution-processed and deposited on diverse substrates to obtain thin films, which has great potential for large-area and flexible electronic devices.^{1–3} Recently, semiconducting polymers for thermoelectric (TE) research have attracted increased attention, aiming at flexible energy generation or heating–cooling devices.^{4–6} The performance of a TE material is related to the figure of merit, $ZT = S^2\sigma T/\kappa$, where S is the Seebeck coefficient, σ is the electrical conductivity, κ is the thermal conductivity, and T is the absolute temperature, respectively. Generally, the pristine semiconducting polymers show high resistance and chemical doping has been effectively used to increase their electrical conductivity.^{6–8} The basic principle of chemical doping is that charge transfer occurs between semiconducting polymers and dopants, generating ionized dopants and (bi)polarons along the backbone of the polymer.⁹ It is known that electrical conductivity is defined by the equation of $\sigma = ne\mu$, where n is the carrier concentration, e is

the elementary charge, and μ is the carrier mobility, respectively. The carrier concentration is originally subjected to the doping efficiency through charge transfer and the separated (bi)polarons being mobile contribute to the electrical conductivity. The doping efficiency, strongly dependent on the driving force for charge transfer, can be improved by choosing a dopant with strong oxidative ability.^{10–12} The carrier mobility is strongly dependent on the morphology and microstructure of the thin film.¹³ The shape and size of dopants inevitably influence the packing of polymers, resulting in different crystalline behaviors and electrical performances. A considerable amount of research has focused on the structure–

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property relationship of semiconducting polymers interacting with 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ), a conjugated small-molecule dopant.^{14–16} Schwartz's group designed a series of dodecaborane (DDB)-based dopants¹⁷ with large size and different shapes, which could effectively shield a doped conjugated polymer from Coulombic interaction with dopant counterions¹⁸ to improve the doping efficiency and carrier mobility. In addition, the molecular weight of polymers also relates to the film mobility associated with the microstructure and connectivity of the crystalline domains.^{13,19–22} Wegner et al. found that P3HT with high molecular weight shows a high degree of crystallinity beneficial to transport properties.²³ Besides organic-conjugated molecules as dopants, inorganic salt-type dopants generally induce higher doping efficiency than that of F4TCNQ because of their stronger oxidation properties.^{24–27} For instance, ferric salts, such as FeCl_3 and Fe(III) trifluoromethanesulfonate, are successfully used to oxidize p-type thiophene polymers.^{24,28,29} The doping process of inorganic salts induces dopant counterions (anions) left into the film interacting with polymer chains. However, how the structure of these anions affects charge transport and electrical conductivity of the polymers has not been well understood. Very recently, the effect of two dopant cations on the electric and thermoelectric properties of an n-doped UFBDDPV polymer was reported.³⁰ It proved that the electrical conductivity of n-type polymers can be improved by precisely modulating the interaction between the cations and the polymer backbone.

Herein, we systematically investigate how the structures of inorganic salt dopants, FeCl_3 , Fe(III) trifluoromethanesulfonate [Fe(OTf)_3], and Fe(III) *p*-toluenesulfonate [Fe(Tos)_3], with different counterions, affect the optoelectronic properties and TE performance of a P3HT film. UV–vis–NIR absorption spectra demonstrate that Fe(OTf)_3 induces higher doping efficiency compared with FeCl_3 and Fe(Tos)_3 under the same conditions. However, the electrical conductivity of a FeCl_3 -doped P3HT film is twice higher than that of a Fe(OTf)_3 -doped film and 10 times higher than that of a Fe(Tos)_3 -doped film. A combination of grazing-incidence wide-angle X-ray scattering (GIWAXS) data, atomic force microscopy (AFM), and Hall-effect measurements indicates that smooth and continuous transport paths existing in a FeCl_3 -doped film contribute to improved carrier transport while discrete domains in the Fe(OTf)_3 -doped film hamper the carrier transport. Therefore, much higher mobility in the FeCl_3 -doped film is observed, which leads to higher electrical conductivity. A prototype device with a five-leg FeCl_3 -doped P3HT film shows a maximum output power of 4.64 nW at the temperature difference of 23.3 K. Our results suggest that the strong oxidation property of dopants is required for charge transfer, but a rational interaction between a dopant counterion and a polymer is much crucial for carrier mobility and thus TE performance.

RESULTS AND DISCUSSION

Energy Level Diagram and Absorption Spectra. The chemical structures of P3HT and Fe(III) salt dopants are shown in Figure 1. We choose P3HT with two different molecular weights, S-P3HT with the number-average molecular weight (M_n) of 14.5 kDa and L-P3HT with M_n of 25.1 kDa determined by gel permeation chromatography (Figure S1). More molecular weight data of P3HT are summarized in Table S1. The HOMO level of a P3HT film was determined

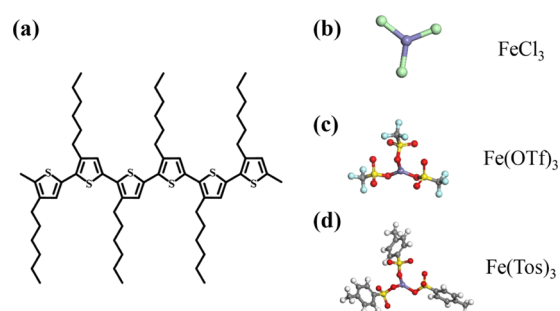


Figure 1. Chemical structures of (a) P3HT polymer and (b–d) iron(III)-salt dopants.

by cyclic voltammetry (CV) (Figure S2). These two batches of the P3HT film show a similar HOMO level around -5.08 eV, although the HOMO level given in the literature is a high-lying value (-4.65 eV) due to different measurement methods.¹⁸ The Fe(III) salt dopants show different LUMO levels determined by CV using a dopant solution. It is found that FeCl_3 shows a high-lying LUMO level located at -4.52 eV compared to that of Fe(OTf)_3 and Fe(Tos)_3 with LUMO levels located at -4.60 and -4.58 eV, respectively. This means Fe(OTf)_3 and Fe(Tos)_3 have stronger oxidative ability than that of FeCl_3 .

Pristine films were prepared by drop-casting the P3HT solution (10 mg mL^{-1} in chlorobenzene) on a glass substrate. After solvent evaporation and annealing at 120°C for 30 min, the films were immersed into the dopant solution (5 mg mL^{-1}) with varied doping times. It should be noted that all dopants were dissolved in a mixed solvent (14 vol % 1-butanol and 86 vol % acetonitrile) to ensure good solubility (Figure S3). To characterize the doped states, the optical absorption of the pristine and doped P3HT films was measured. As shown in Figure 2, pristine S-P3HT and L-P3HT films show a similar absorption spectra range and no absorption over 700 nm is observed. Upon doping, the absorption feature of the doped P3HT film is quite different after immersion in two different dopant solutions for 30 min. We assigned two absorption bands, band_i: $320\text{--}700 \text{ nm}$ and band_{II}: $700\text{--}2500 \text{ nm}$, to neutral polymer and (bi)polaronic absorption, respectively. It is found that the integrated area ratio (R) of two bands (band_{II}/band_i) for the Fe(OTf)_3 -doped P3HT film is two times higher than that of the FeCl_3 -doped film without batch-to-batch variation (Table S2). This indicates a higher doping level for the Fe(OTf)_3 -doped P3HT film under the same conditions, which could be attributed to the lower LUMO level of Fe(OTf)_3 , endowing a larger driving force for charge transfer (Figure 2a). However, the doping level of Fe(Tos)_3 -doped P3HT is the lowest, although Fe(Tos)_3 shows the similar LUMO level with Fe(OTf)_3 . This indicates that the charge-transfer process requires the insertion of dopants into a polymer network and large size of *p*-toluenesulfonate anion could induce steric hindrance and thus reduce the effective oxidation of a polymer backbone.¹⁴ Raman spectroscopy was employed to verify the doping process (Figure S4). We focused on the modes, C=C symmetric stretching at 1452 cm^{-1} , and the C–C intraring stretching at 1386 cm^{-1} , which are sensitive to π -electron delocalization and degree of structural order in poly(thiophenes).³¹ The Raman spectra were normalized by the intensity of the C=C vibrational mode. A new low-energy wing assigned to the C=C symmetric stretching peak appears after doping, arising from

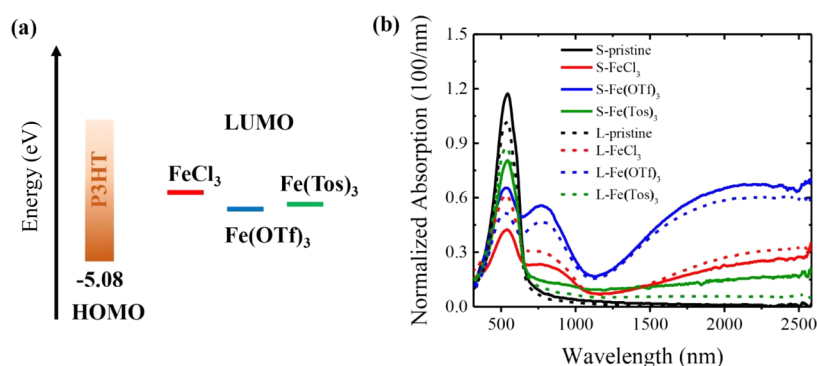


Figure 2. (a) Energy diagram of the P3HT and dopants. (b) Thickness-normalized UV–vis–NIR absorption spectra for undoped and doped P3HT. P3HT films with small molecular weight: solid lines; P3HT films with large molecular weight: dash lines. All doped films were immersed in a dopant solution for 30 min.

the increased conformational order of polymer chains, which indicates the formation of quinoid units of oxidized P3HT. The peak of the Fe(OTf)₃-doped film shows the lowest energy compared with that of two other dopants, meaning the highest doping level Fe(OTf)₃-doped film being in agreement with the absorption spectra. It is also observed that the peaks at 1386 cm⁻¹ become broadened upon doping for all of these films, which can be attributed to the conformational disorder after adding dopants. This peak is obviously broadened for Fe(Tos)₃-doped P3HT, suggesting a large amount of disordered structures in the film due to its large size. These comparisons indicate that the formation of (bi)polarons is synergistically affected by the energy level and interaction between a polymer and a dopant in the microstructure.

Electrical Conductivity of Doped P3HT. We then analyze the effect of doping time on electrical conductivity of the P3HT film. In Figure 3, a higher electrical conductivity can

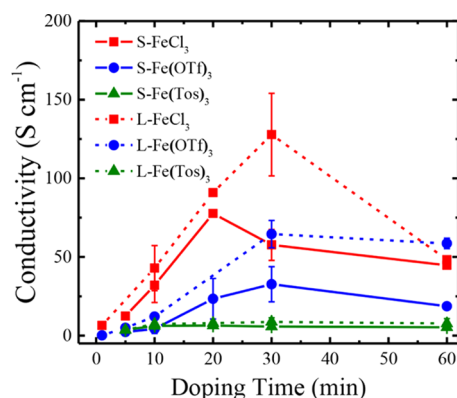


Figure 3. Electrical conductivities of FeCl₃, Fe(OTf)₃, and Fe(Tos)₃-doped P3HT films as a function of the doping time. P3HT films with small molecular weight: solid lines, P3HT films with large molecular weight: dash lines. Each data point at 30 min is an average of at least four devices.

be obtained for L-P3HT compared to that of S-P3HT, which is reasonable due to a longer effective π -conjugation in the L-P3HT film for charge transport. It is observed that the electrical conductivity of FeCl₃-doped P3HT is significantly higher than that of Fe(OTf)₃ and Fe(Tos)₃-doped films for both S-P3HT and L-P3HT. For instance, the electrical conductivity of FeCl₃-doped L-P3HT (128 S cm⁻¹) is 2 times higher than that of Fe(OTf)₃-doped L-P3HT (65 S

cm⁻¹) and 10 times higher than that of Fe(Tos)₃-doped L-P3HT (9 S cm⁻¹), respectively, doping for 30 min, although a higher doping level is observed for the Fe(OTf)₃-doped film (Figure 2b). In addition, over-doping, occurring for these films immersed in a dopant solution for 60 min, can result in a significant drop of electrical conductivity. All of these films show similar Seebeck coefficients immersed in a dopant solution for 30 min (Figure S4). Combining the electrical conductivity and Seebeck coefficient data, the power factor of FeCl₃-doped L-P3HT (20 μ W m⁻¹ K⁻²) is 4 times higher than that of Fe(OTf)₃-doped L-P3HT (5 μ W m⁻¹ K⁻²) and 6 times higher than that of Fe(Tos)₃-doped L-P3HT (3 μ W m⁻¹ K⁻²). The temperature-dependent electrical conductivity, Seebeck coefficient, and power factor for FeCl₃-doped L-P3HT are shown in Figure S6. It is observed that a highest power factor could be obtained around 340 K. The output voltage and output power versus current were measured for FeCl₃-doped L-P3HT using the homemade apparatus (Figure S7). At the temperature difference of 23.3 K, the measured maximum output power is about 4.64 nW, which indicates the potential application of this film after further experimental optimization.

With the different doping levels of three dopants in consideration, we propose that most of (bi)polarons in Fe(OTf)₃-doped films could not contribute to the electrical conductivity.³² This indicates that the carrier mobility associated with the solid-state packing would be a key contributor to their huge difference in electrical conductivity. We next investigate the mobile carrier in the doped L-P3HT film by employing Hall-effect measurements (Figure S8). We notice that it is difficult to extract reliable data for Fe(Tos)₃-doped P3HT due to too low carrier concentration in the film, so we just focus on the comparison of other two dopants. As summarized in Table S3, the mobile carrier concentration follows the same trend, as in absorption spectra in Figure 2b. The mobility of FeCl₃-doped P3HT is much higher than that of Fe(OTf)₃-doped P3HT. This indicates that the high electrical conductivity is mainly contributed by the high mobility, possibly ascribed to the superior microstructure for charge transport in FeCl₃-doped P3HT.

Microstructure Characterization of the Thin Films.

The crystallinity property of the pristine P3HT and doped films (immersed into a dopant solution for 30 min) were analyzed by GIWAXS (Figure 4) and all data were summarized in Table S4. In the out-of-plane pattern (q_z) for pristine and doped S-P3HT films, multiple-order lamellar stacking (h00) peaks were observed. In addition, a lamellar stacking (100)

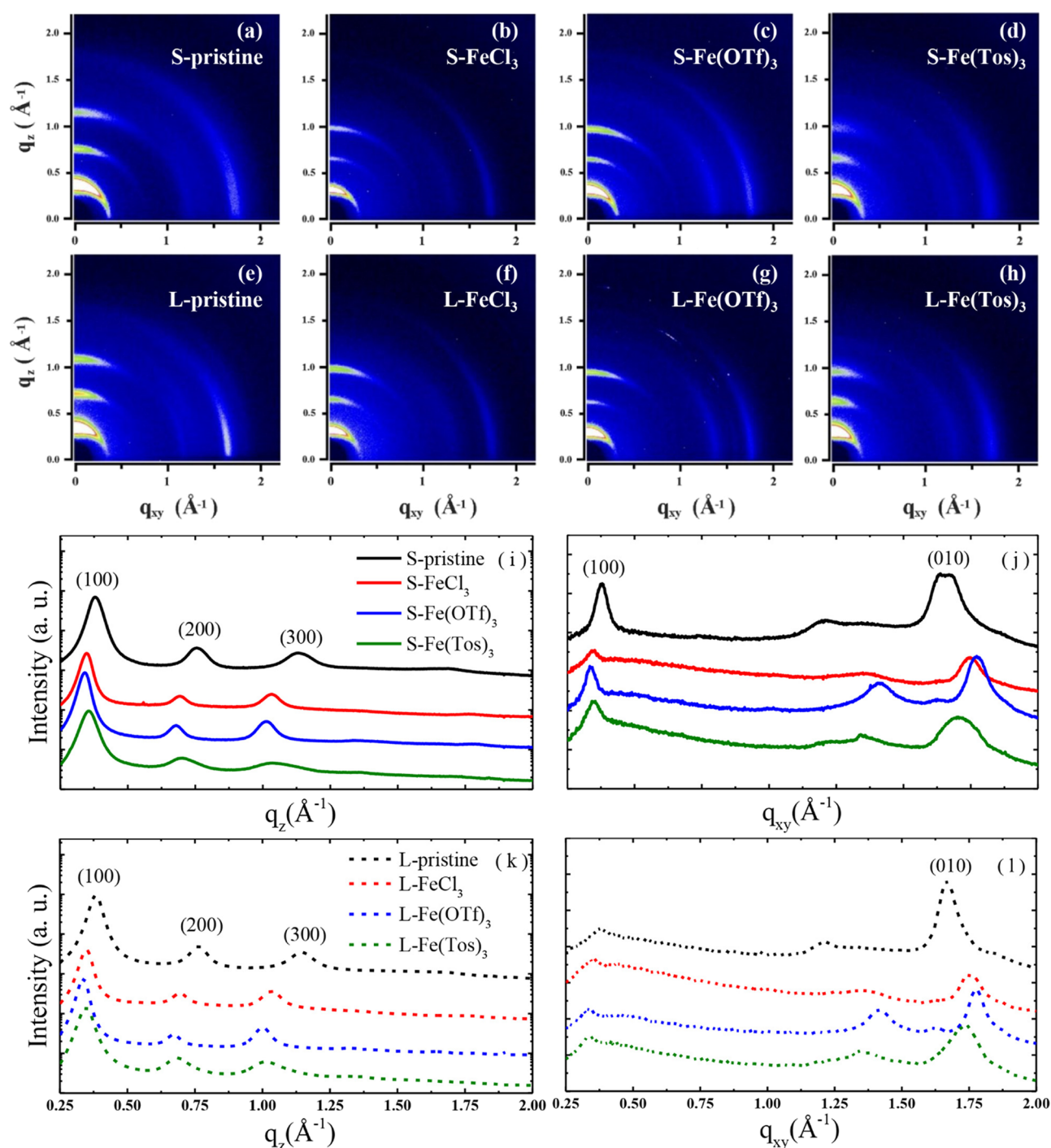


Figure 4. (a–h) 2D GIWAXS images of pristine P3HT and doped P3HT films. (a, e) Pristine P3HT films, (b, f) FeCl₃-doped P3HT films, (c, g) Fe(OTf)₃-doped P3HT films, and (d, h) Fe(Tos)₃-doped P3HT films. (i–l) GIWAXS line cuts of out-of-plane scattering and in-plane scattering. (i, k) Out-of-plane scattering profiles and (j, l) in-plane scattering profiles. P3HT films with small molecular weights: solid lines, P3HT films with large molecular weights: dashed lines. All doped films were immersed in a dopant solution for 30 min.

peak also appears along the in-plane (q_{xy}) direction, indicating the coexistence of edge-on and face-on orientations in films. In contrast, the dominated edge-on orientations are observed in all L-P3HT films. On the other hand, the lamellar distance and π - π stacking distance of pristine S-P3HT are 16.59 and 3.79 Å, respectively, compared with those of 16.41 and 3.77 Å, respectively, for L-P3HT. These results support the fact that the high molecular weight of the polymer is beneficial for highly ordered and intense packing.²⁰ Upon doping for 30 min, the lamellar distance of P3HT increases while the π - π stacking

distance decreases for all films (Table S4), suggesting that the dopants prefer to intercalate into side chains.³³ It is found that Fe(OTf)₃-doped P3HT shows the shortest π - π stacking distance and the longest lamellar distance compared with those of FeCl₃ and Fe(Tos)₃-doped P3HT both observed in two batches of P3HT. The lamellar distance is smallest and π - π stacking is largest for Fe(Tos)₃-doped P3HT among three dopants, probably attributed to the lowest doping efficiency due to its large size of the dopant anion. The trends for lamellar and π - π stacking distances of these doped films are in

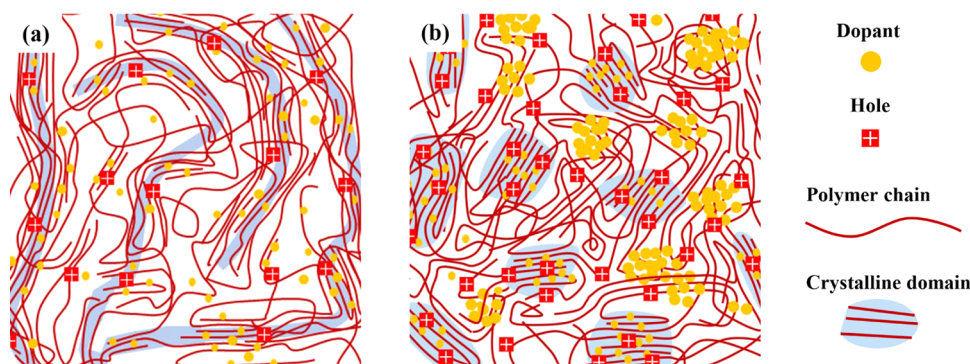


Figure 5. Schematic illustrations of the microstructures of the P3HT films doped by (a) FeCl_3 and (b) $\text{Fe}(\text{OTf})_3$.

good agreement with the doping levels shown in Figure 2b. The higher doping level means stronger coupling of oxidized polymer chains, which results in a shorter π – π stacking distance accompanied with a larger lamellar distance due to efficient intercalation of dopants into side chains. It is generally accepted that the charge transport is dominantly limited by interchain charge hopping, and the short π – π stacking distance is desired to contribute to high carrier mobility.^{34,35} However, higher mobility was obtained for FeCl_3 -doped P3HT in our work compared to the mobility of $\text{Fe}(\text{OTf})_3$ -doped P3HT. This indicates that the high-doped regions in the $\text{Fe}(\text{OTf})_3$ -doped film are probably discrete, resulting in a discontinuous charge transport path. To further understand the detailed microstructure in the films, the full-width at half-maximum (FWHM) was calculated from the scattering peak (100) to compare the size of the crystalline grain (Table S4).^{36,37} It is observed that the crystalline grain of P3HT is enlarged after doping with FeCl_3 and $\text{Fe}(\text{OTf})_3$, whereas $\text{Fe}(\text{OTf})_3$ -doped P3HT has a smaller FWHM, corresponding to a larger crystalline size in the film compared with that of FeCl_3 -doped P3HT. The increase of the crystalline grain indicates that the introduction of $\text{Fe}(\text{OTf})_3$ significantly improves polymer packing and film crystallinity while FeCl_3 molecules moderately intercalate into side chains of P3HT with slight disruption of polymer packing. In contrast, the crystalline size of P3HT is slightly reduced after doping with $\text{Fe}(\text{Tos})_3$, which suggests the absence of long-range conformational ordering induced by charge transfer between a polymer and a dopant,³⁸ in accordance with the low doping level, as shown in Figure 2b. The microstructure of these films was further depicted by the analysis of surface morphology determined by atomic force microscopy (AFM) (Figure S9). It is found that the FeCl_3 -doped P3HT film is smooth due to good miscibility between polymers and dopants. However, some large crystallites from dopant self-aggregation are observed in the $\text{Fe}(\text{OTf})_3$ -doped film, in particular, in the L-P3HT film, which could be attributed to the intense packing of L-P3HT so that the dopants are hard to insert into the polymer chains. For the $\text{Fe}(\text{Tos})_3$ -doped film, it seems that dopants intend to precipitate and stay at the film surface. The obvious difference of the dopant-dependent film morphology supports that the continuous interconnected network is beneficial for mobility of the FeCl_3 -doped P3HT film and the discrete crystallites in the $\text{Fe}(\text{OTf})_3$ -doped P3HT film hamper the carrier transport.

Combining the results of GIWAXS with Hall-effect experiments, we propose the model of doping process in these films. Dopant molecules tend to intercalate into the space of P3HT side chains, and the doping process in a dopant solution

accompanying film swelling induces much strong π – π stacking of the polymer backbone and leaves a dopant anion into the polymer network. For FeCl_3 , the good miscibility between polymers and anions (FeCl_4^- or Cl^-) results in a homogenous morphology for continuous carrier transport paths (Figure 5a). In contrast, for $\text{Fe}(\text{OTf})_3$, ionized P3HT chains strongly aggregate together and form large crystalline domains along with dopants that are partially self-aggregated (Figure 5b). The heterogeneity of the film with discrete crystalline regions hampers the carrier transport due to carriers moving from one conjugated backbone to another at places where conjugation is interconnected. In addition, low-crystallinity S-P3HT shows the same phenomenon in GIWAXS and AFM, indicating that the effect from dopant anions is independent of both molecular weight and intrinsic crystallinity of the P3HT polymer.

To further support our charge transport model above, the temperature-dependent electrical conductivity of FeCl_3 and $\text{Fe}(\text{OTf})_3$ doped P3HT was investigated, as shown in Figure 6.

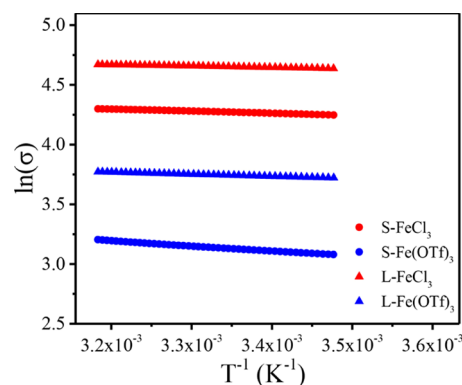


Figure 6. Temperature-dependent electrical conductivity of doped P3HT films (the unit of σ is S cm^{-1}). All doped films were immersed in a dopant solution for 30 min.

It is found that the charge carrier transport follows the thermal activation hopping mechanism.^{39,40} The activation energy was calculated according to the Arrhenius equation: $\sigma = \sigma_0 \exp(-E_a/k_B T)$, where σ_0 is a temperature-independent prefactor and k_B is the Boltzmann constant.⁴¹ For two different dopants, S-P3HT films show much higher E_a values than those of L-P3HT films attributed to the lower crystallinity in the S-P3HT film. Importantly, it is found that FeCl_3 -doped S-P3HT shows a lower E_a of 16.8 meV than that of a $\text{Fe}(\text{OTf})_3$ -doped film ($E_a = 40.5$ meV), as well as for L-P3HT ($E_a = 11.0$ meV for FeCl_3 -doped L-P3HT and $E_a = 16.6$ meV for $\text{Fe}(\text{OTf})_3$ -doped L-P3HT, respectively), meaning that a smaller transport barrier

exists in the FeCl_3 -doped film. This is in good agreement with the microstructure character illustrated in Figure 5. As we discussed above, the formation of (bi)polarons is strongly dependent on the driving force for charge transfer, therefore, the strong oxidative ability of dopants is required during choosing dopants. However, the interaction between a polymer chain and dopant counterions in the microstructure seems to be the key for the charge transport and thus for improving the electrical conductivity. The size and shape of dopant counterion matching with the structure characteristic of the polymer should be synergistically considered during the structure design toward high TE performance.

CONCLUSIONS

We have investigated P3HT films doped by three iron (III) salts with different anions. It is found that the doping level strongly depends on the driving force for charge transfer and is also limited by the insertion of the dopant into the polymer network. For the $\text{Fe}(\text{OTf})_3$ -doped film, despite the higher doping level, its electrical conductivity is lower than that of the FeCl_3 -doped film. This can be attributed to higher carrier mobility in the FeCl_3 -doped film determined by Hall-effect measurements. GIWAXS measurements give evidence that a much stronger π - π stacking of the P3HT backbone and larger crystalline domains exist in the $\text{Fe}(\text{OTf})_3$ -doped film compared with those of FeCl_3 -doped P3HT. Moreover, smaller activation energy for carrier transport is obtained for the FeCl_3 -doped film, demonstrating lower carrier transport barriers in this film. As a consequence, it could be concluded that continuous carrier transport paths form in the FeCl_3 -doped film are responsible for high mobility while discrete domains in the $\text{Fe}(\text{OTf})_3$ -doped film hamper the carrier transport, in accordance with the surface morphology, as shown in AFM images. For a dopant with a large anion, $\text{Fe}(\text{Tos})_3$, both the doping efficiency and long-range order of the polymer are inhibited although it has strong oxidation ability. A prototype device with a five-leg FeCl_3 -doped L-P3HT shows a maximum output power of 4.64 nW at the temperature difference of 23.3 K, indicating a potential application of these materials at ambient conditions. Our results suggest that the size and shape of dopant counterions affecting the microstructure are crucial for high electrical conductivity except for taking the oxidation ability into account. The design of a polymer-dopant system should combine the energy level and miscibility together for achieving high thermoelectric performance.

EXPERIMENTAL SECTION

Materials and Characterization. Poly(3-hexylthiophene) (P3HT, Aldrich), chlorobenzene (CB, Alfa-Aesar, 99.5%), iron(III) chloride (FeCl_3 , Acros, 98%), iron(III) trifluoromethanesulfonate ($\text{Fe}(\text{OTf})_3$, Damas- β , 95%), iron(III) *p*-toluenesulfonate hexahydrate ($\text{Fe}(\text{Tos})_3 \cdot 6\text{H}_2\text{O}$, Aldrich, technical grade), acetonitrile (CH_3CN , >99.5%, Alfa), and 1-butanol (>99.9%, Alfa) were used. All materials were used as obtained without further purification. Gel permeation chromatography (GPC) was conducted by Polymer Laboratories PL-GPC220. Cyclic voltammetry data were taken on a CH-Instruments 600E Electrochemical Workstation. A three-electrode system was used: Ag/Ag^+ (saturated KCl in CH_3CN as a fill solution), a platinum wire, and a platinum plate used as a reference electrode, a counter electrode, and a working electrode, respectively. The supporting electrolyte was 0.1 M tetrabutylammonium hexafluorophosphate (Bu_4NPF_6) dissolved in anhydrous acetonitrile. Ferrocene/ferrocenium (Fc/Fc^+) was used as external calibration. UV-vis-NIR absorption spectra were measured on HITACHI U-4100. Raman

spectra were measured with a Renishaw inVia Reflex confocal Raman microscope. Grazing-incidence wide-angle X-ray scattering (GIWAXS) data were recorded at beamline BL14B1 of the Shanghai Synchrotron Radiation Facility (SSRF) at a wavelength of 1.24 Å. Atomic force microscopy (AFM) was performed using NT-MDT NTEGRA. Electrical conductivity and the Seebeck coefficient were measured with a ZEM-5 TF (ADVANCE RIKO, Inc.). Film thicknesses were measured using a Bruker DektakXT contact profilometer. The Hall effect of samples was measured using a LakeShore CRYOTRONICS 8400 series system (TOYO Corporation) with the AC field Hall method with a frequency of 100 mHz. The gold electrode layers were deposited under high vacuum ($<1 \times 10^{-5}$ Pa) with thicknesses of 500 Å. Temperature-dependent conductivity was measured using a physical property measurement system (PPMS, Quantum Design).

Sample Preparation and Doping. Glass slides were cleaned by sequential sonication steps in deionized water, acetone, and isopropyl alcohol, and then they were dried and stored in a sealed box before use. P3HT was dissolved in CB with a concentration of 10 mg mL^{-1} and it was stirred and heated on a hotplate at 50°C for 6 h before fabrication by drop-casting this solution onto glass substrates. P3HT thin films were annealed at 120°C for 30 min in an argon glovebox (H_2O , $\text{O}_2 < 0.1 \text{ ppm}$) and then cooled to room temperature. FeCl_3 , $\text{Fe}(\text{OTf})_3$, and $\text{Fe}(\text{Tos})_3$ were, respectively, dissolved in a mixed solvent (the volume ratio is 14 vol % 1-butanol to 86 vol % acetonitrile) with a concentration of 5 mg mL^{-1} . The doping process was performed by immersing the film individually into a dopant solution and then the films were rinsed with dry CH_3CN quickly to remove excess dopants on the surface. All steps were carried out under ambient conditions. For samples used for Hall-effect experiments, gold electrodes of 50 nm thickness were deposited on the annealed P3HT films under ultrahigh vacuum. Then, the films were immersed into a dopant solution for 30 min and rinsed with dry CH_3CN .

Fabrication and Measurement of the Thermoelectric Module. The doped films used for Hall-effect experiments were directly employed to fabricate the TE module. First, five pieces of the FeCl_3 -doped L-P3HT film were fixed and arranged in parallel. Then, each film was electrically connected in a series by Pt wires fastened by painting the Ag paste at both sides. The hot end of the prototype module was placed on a copper plate connected with a voltage adjustable heating slice to control its temperature. The cold end of the module was placed on a metal block. Performance measurement of the prototype module was conducted using a homemade apparatus. The temperature difference between the hot end and the cold end was set up to be 23.3 K. Series data of the output voltage (V) and current (I) were measured by adjusting the load resistance. The output voltage could be obtained by the equation: $V = I \times R_{\text{load}} = V_{\text{OC}} - I \times R_{\text{in}}$, where R_{load} is the external resistance, V_{OC} is the open-circuit voltage, and R_{in} is the internal resistance, respectively. When R_{load} approximately equals to the internal resistance, the measured maximum output power is about 4.64 nW, which could be calculated by $P = VI = V_{\text{OC}} \times I - I^2 \times R_{\text{in}}$.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsaelm.0c01067>.

GPC curves, the cyclic voltammogram, Raman spectra, thermoelectric parameters, AC-Hall data, and AFM images (PDF)

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